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Multilingual Dependency Parsing of Uralic Languages

Parsing with zero-shot transfer and cross-lingual models using geographically proximate, genealogically related, and syntactically similar transfer languages

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Abstract

One way to improve dependency parsing scores for low-resource languages is to make use of existing resources from other closely related or otherwise similar languages. In this paper, we look at eleven Uralic target languages (Estonian, Finnish, Hungarian, Karelian, Livvi, Komi Zyrian, Komi Permyak, Moksha, Erzya, North Sámi, and Skolt Sámi) with treebanks of varying sizes and select transfer languages based on geographical, genealogical, and syntactic distances. We focus primarily on the performance of parser models trained on various combinations of geographically proximate and genealogically related transfer languages, in target-trained, zero-shot, and cross-lingual configurations. We find that models trained on combinations of geographically proximate and genealogically related transfer languages reach the highest LAS in most zero-shot models, while our highest-performing cross-lingual models were trained on genealogically related languages. We also find that cross-lingual models outperform zero-shot transfer models. We then select syntactically similar transfer languages for three target languages, and find a slight improvement in the case of Hungarian. We discuss the results and conclude with suggestions for possible future work.

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1 Introduction

1.1 Overview

The expansion of the field of natural language processing (NLP) in recent years has allowed for more and more NLP tasks to be carried out with greater efficiency. However, at the heart of any NLP research lies any number of data, and despite recent advancements, manually annotating natural language data remains an expensive and time-consuming task.

Of the many languages that exist in the world, annotated data created for NLP tasks only exist for a very small percentage. For many of the languages in that percentage, only a very small amount of data exist. Generally speaking, having more data available for a language will mean more tasks and more accurate results will be possible for that language, while other languages do not have that same benefit.

Low-resource (LR) languages, i.e. languages with little to no annotated data, have been the subject of plenty of previous studies focused on variations of the same question: how can we apply NLP tasks to these languages with the small amount of resources, or complete lack thereof, available to us?

In this paper we will be focusing on one particular NLP task: dependency parsing. We will be using a transfer learning approach to boost parsing accuracy in LR Uralic languages. In transfer learning, the resources from a transfer language are used to support the processing of a target language. In the case of LR target languages, transfer learning therefore comes with the massive benefit of allowing us to make use of the resources from high-resource (HR) transfer languages.

The choice of transfer languages is vital in determining how efficiently transfer learning will boost parsing, and choosing these languages is not always straightforward. Two given languages could be close in various ways; they could, for example, be genealogically related and be classified as belonging to the same language family, or they could be from entirely different families but still share important syntactic similarities. When choosing transfer languages for parsing, we therefore need to decide what type of language distance is the most important to consider.

Once transfer languages have been selected, there is also the question of how they should be used; is it best to train a model with just one language, or to combine multiple languages? Is there a significant difference in performance when training a model with just a combination of transfer languages, vs. adding what little data we can from the LR target language?

1.2 Purpose and Motivation

The purpose of this paper is as follows:

- To select potential transfer languages for a selection of Uralic target languages, based on three different types of language distances, i.e. geographic proximity, genealogical relatedness, and syntactic similarity;

- To determine which of these three categories of potential transfer languages (or combination thereof) is the most useful basis for transfer language selection for dependency parsing;
- To determine which of four different model configurations (target-trained, single-source zero-shot, multi-source zero-shot, and cross-lingual) yield the best results when testing target languages.

While we cannot necessarily transfer exact results to other languages not included in this study, the results may give us some understanding of the overall effects of training models on various combinations of transfer languages (varying in similarity to the target language) and data from the target languages themselves.

1.3 Outline

In the remainder of this paper, we will first explore related work which inspired this study in Chapter 2. Here, we will discuss other work on transfer learning, choosing transfer languages, specific ways to combine transfer languages and structure models, and more.

In Chapter 3, we will discuss the methodology, including the dependency parser and data used. This chapter also includes the detailed description of the different types of models trained in the 'Experimental Setup' section, as well as the evaluation metrics.

In Chapter 4, the results are presented and discussed.

In Chapter 5, the paper is concluded and possibilities for future work are discussed.

2 Background

2.1 Dependency Parsing

Dependency parsing gives us information about the structures of sentences in a text by identifying and labeling the head and the dependent in a pair of words (Kubler et al., 2009). For example, in the sentence "I speak", *speak* is the head, and *I* is the dependent. In this particular dependency arc, the dependent *I* is the subject of the sentence.

The parse tree of a more complex sentence can be seen in Figure 2.1.

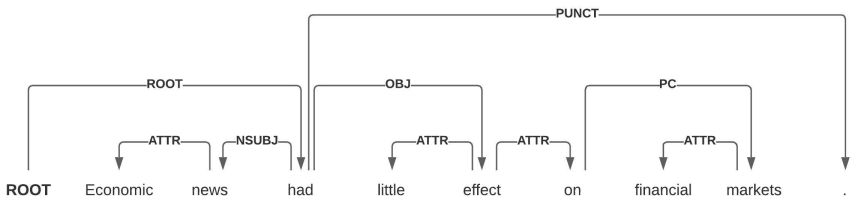

Figure 2.1

In this figure, we can see several different types of dependencies; *news* and *effect*, for example, are dependent on the verb *had*, with the dependency types *subject* and *object*, respectively. *Little* is a dependent of type *attribute* of the head *effect*, and so on; each word and punctuation mark in the sentence has a head and/or dependent, and each of these dependency relationships is marked with a specific dependency type.

There are certain rules that parse trees must adhere to. Note how the words *had* and *effect* have multiple dependents; the opposite, wherein a dependent has multiple heads, is not possible.

This type of parsing is extremely useful when used as part of more complex NLP tasks, such as machine translation, and thus developing methods to effectively parse LR languages is crucial in order to be able to apply these tasks to said languages.

2.1.1 Transition-based Parsing

Several methods of dependency parsing have been proposed. One commonly used method is known as transition-based parsing (Nivre, 2003; Gómez-Rodríguez and Nivre, 2010; Attardi, 2006).

In transition-based parsing, the algorithm applies certain actions (transitions) to move from one configuration of partially processed sentences to another. The algorithm uses a stack, S, containing the partially processed tokens, a buffer, B, containing the tokens yet to be processed, and a set of labeled arcs, A, applied to dependencies.

During the parsing process, the first word from the buffer is moved to the stack. If there is a dependency between this word and the next word in the buffer, it is applied here and added to the set of arcs. The type of arc can differ; if the head is in the stack, it is a right-arc, and if it is in the buffer it is a left-arc. In the case of a left-arc transition, the dependent is removed from the stack. In the case of a right-arc transition, the head remains in the stack, as a head can have multiple dependents, meaning it is not

necessarily fully processed yet. Only once tokens are fully processed are they removed from the stack.

Transitions continue to be applied in this manner until every single token has been processed and the set of generated arcs creates a full parse tree of the input sentence.

As an example, for the sentence "He worked for the BBC for a decade," we might see the following steps in a transition-based algorithm:

Transition	B	S	A
	([ROOT]	[He, ...]	0)
$SH \rightarrow$	([ROOT, He]	[worked, ...]	0)
$LA_{nsubj} \rightarrow$	([ROOT]	[worked, ...]	A_1 = {worked, nsubj, He})
$RA_{root} \rightarrow$	([ROOT, worked]	[for, ...]	$A_2 = A_1 \cup$ {ROOT, root, worked})
$SH \rightarrow$	([ROOT, worked, for]	[the, ...]	A_2)
$SH \rightarrow$	([ROOT, worked, for, the]	[BBC, ...]	A_2)
$LA_{det} \rightarrow$	([ROOT, worked, for]	[BBC, ...]	$A_3 = A_2 \cup$ {BBC, det, the})
$LA_{case} \rightarrow$	([ROOT, worked]	[BBC, ...]	$A_4 = A_3 \cup$ {BBC, case, for})
$RA_{obl} \rightarrow$	([ROOT, worked, BBC]	[for, ...]	$A_5 = A_4 \cup$ {worked, obl, BBC})
$REDUCE \rightarrow$	([ROOT, worked]	[for, ...]	A_5)
$SH \rightarrow$	([ROOT, worked, for]	[a, decade, .]	A_5)
$SH \rightarrow$	([ROOT, worked, for, a]	[decade, .]	A_5)
$LA_{det} \rightarrow$	([ROOT, worked, for]	[decade, .]	$A_6 = A_5 \cup$ {decade, det, a})
$LA_{case} \rightarrow$	([ROOT, worked]	[decade, .]	$A_7 = A_6 \cup$ {decade, case, for})
$RA_{obl} \rightarrow$	([ROOT, worked, decade]	[.]	$A_8 = A_7 \cup$ {worked, obl, decade})
$REDUCE \rightarrow$	([ROOT, worked]	[.]	A_8)
$RA_{obl} \rightarrow$	([ROOT, worked, .]	[]	$A_9 = A_8 \cup$ {worked, punct, .})

The full parse tree for this sentence is depicted in Figure 2.2.

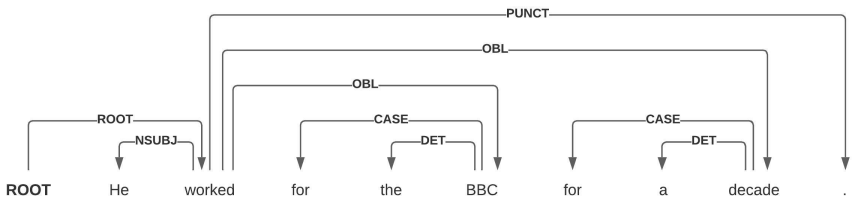

Figure 2.2

UUParser

Some earlier uses of transition-based parsing dealt exclusively with projective dependency structures (Nivre, 2003). In a projective structure, the arcs denoting a dependency between two words do not overlap with other such arcs between two other words in a sentence. A non-projective structure contains overlapping arcs.

Various methods have been proposed for parsing non-projective structures (Gómez-Rodríguez and Nivre, 2010; Attardi, 2006; Nilsson and Nivre, 2005).

UUParser is a transition-based dependency parser developed by de Lhoneux et al., 2017. This parser uses a method based on Nivre, 2009, to reorder words in non-projective structures by adding an extra transition to the parsing algorithm. They combined this method with the arc-hybrid system proposed by Kuhlmann et al., 2011, and found they were able to parse non-projective structures with good results.

They also introduced the concept of a static-dynamic oracle. This oracle functions dynamically with the *shift*, *left*, and *right* transitions, but statically with the transition they added to reorder non-projective structures (called *swap*). The use of this oracle allows the parser to learn from non-optimal transitions during the training process.

2.2 Multilingual Parsing

In order to parse LR languages, we can make use of the existing resources from other languages.

One approach for multilingual parsing is projection annotation. Some early studies (Hwa et al., 2002; Hwa et al., 2004) demonstrate how even very cheap model transfer can improve upon baseline scores by projecting English structures directly onto other languages using parallel texts.

Many studies focusing on projection annotation make use of parallel texts, which are extremely useful when available; however, for the thousands of languages in the world, large corpora like these are difficult to come by. Exceptions are religious texts such as the Bible and the Watchtower, which provide parallel texts in hundreds of languages. Agić et al., 2016, made use of these texts; they projected edge scores from multiple source languages and found that this yielded a significant improvement in the parsing of LR languages specifically.

Relying on parallel texts limits which languages we can apply NLP tasks to, however, as these types of resources are still not available in all languages, and creating them from scratch is expensive.

2.2.1 Model Transfer

Another approach is to train a parsing model with existing data from languages other than the target language (Smith et al., 2018; Meechan-Maddon and Nivre, 2019). Unlike with projection annotation, model transfer does not require parallel texts, making it easier to use in cases where these texts are not available. This also means we do not need to rely on the types of texts for which parallel translations typically are widely available, such as the Bible or the Watchtower. Instead, we can make use of data from a wider variety of sources.

Ammar et al., 2016, trained one model to parse a variety of target languages, noting the biggest improvement among those with less or no training data.

Lim et al., 2018, manually annotated a handful of sentences as the starting point of their experiments. They also found that using transfer languages that are closely related to the target language is far more useful than using those that are not related.

2.3 Choosing Transfer Languages

Smith et al., 2018, found that it is beneficial to make use of closely related transfer languages rather than unrelated ones.

Meechan-Maddon and Nivre, 2019, presented a series of experiments with different types of models using combinations of target and transfer training data. Their findings suggest that training a model on both training and transfer data often yields higher results than training data alone. This is especially useful when not much training data is available, though does rely on the availability of data for closely related languages.

One of the target languages that they looked at was North Sámi, and they found that using Finnish, Estonian, and Hungarian as transfer languages for North Sámi yielded promising results.

2.3.1 Types of Language Similarity

Lin et al., 2019, also acknowledged the benefits of using closely related transfer languages, but point out that these languages are often chosen based on the researchers' intuition about the distances between languages. They present a method for choosing transfer languages based on more varied criteria, including syntactic and geographic distance, suggesting that taking all of these factors into account is far superior to only using the most closely related languages.

Based on their findings, Lin et al., 2019, also presented the order of importance of various types of language distances for specific NLP tasks. They considered the effect of different distance features on parsing when choosing transfer languages. They looked at both dataset-dependent and dataset-independent features.

Dataset-dependent features are derived from the actual datasets rather than the language as a whole, which means that these features are only relevant when big enough datasets are established for each language. Thus, they are not ideal when focusing on LR languages. We will therefore focus on dataset-independent features, which are derived from the linguistic study of languages, rather than specific datasets.

Lin et al., 2019, found that in the case of dependency parsing, the top three most useful types of dataset-independent language distance features are geographic, genealogical, and syntactic distance. Their experiments were limited to zero-shot transfer models, i.e. models were trained only on data from the transfer languages and not on any data from the target language.

In the case of Uralic languages, there are therefore a few different transfer languages from other language families to consider; several Uralic languages are spoken within parts of Russia and have been affected by their close geographic distance to Russian, while others are spoken further West, where they have had more contact with Scandinavian languages. Lim et al., 2018, looked at Komi-Zyrian as a target language, and found that aside from other Uralic languages, Russian was a useful transfer language.

2.4 Proxy Treebanks

Stymne et al., 2018, introduced a treebank embedding to train single-source models on multiple treebanks. This embedding combines all available treebanks for a language to create a single model. When testing a treebank, this method allows for that treebank's corresponding embedding to be used, improving parsing scores because the differences between treebanks, such as sizes, annotation choices, and genres, are taken into account. For example, if one treebank used in training a model uses a different annotation scheme than the rest, this method ensures that the treebank embedding used when testing that same treebank is from that treebank only.

One issue that can arise when working with multilingual parsing is that in some cases, test data from a given treebank will not have any available training set. In other words, a treebank we may want to test with different models will not have been used to train those models, meaning there will not be a treebank embedding for that treebank available.

To solve this issue, Stymne et al., 2018, introduced what they call a proxy treebank. Simply put, a proxy treebank is the treebank embedding from a different treebank (one that was used for the training process) when the test set does not have a corresponding embedding of its own. This technique has also been used in other research, such as Smith et al., 2018.

Wagner et al., 2020, took this idea one step further by introducing interpolated treebank embedding vectors, which further improved parsing performance for languages not included in training a model.

3 Methodology

3.1 Parser

The parser used for the experiments, UUParser version 2.3, was developed by de Lhoneux et al., 2017.

This parser has been used in papers such as Meechan-Maddon and Nivre, 2019, with promising results for languages including the Uralic language of North Sámi.

The default settings of the parser were used throughout. The existing tokenization from UD (Nivre et al., 2019), is used for all data, and POS tags are not used.

3.2 Uralic Target Languages

Of the 11 Uralic languages included in Universal Dependencies 2.5 (Nivre et al., 2019) not all are LR languages. The treebanks used for each language vary significantly in size, from Komi Permyak, with just 71 sentences, to Estonian, with around 30,000. The size of each treebank can be seen in Table 3.1.

Another important distinction between the different treebanks is that not all of them have training sets. The Finnish, Estonian, Hungarian, North Sámi, and Livvi treebanks have training sets and are therefore used to train models, while the other six treebanks only have test sets, and are therefore used strictly for testing.

Out of the treebanks that have training sets, two do not have dev sets: those for North Sámi and Livvi. In order to obtain dev data when training models with treebanks that do not have dev sets, we can use the UUParser flag *create-dev* to randomly select a certain percentage of the existing data to use as a dev dataset. We can make use of this flag for North Sámi. However, the Livvi treebank is too small for this option to work, and so we will not have any dev data for Livvi.

While all 11 languages are part of the Uralic language family, they are not equally closely related to each other. The language family can be subdivided into five different branches, which are shown in Table 3.1.

Of the 11 languages, all but four use some form of the Latin alphabet. The remaining four, which use the Cyrillic alphabet, are Komi Zyrian, Komi Permyak, Moksha, and Erzya.

3.3 Transfer Languages

As previous research has suggested (Lin et al., 2019, Lim et al., 2018), it is useful to consider transfer languages with low distances to the target languages in different distance categories. We will look at transfer languages close to the target languages in the top three most important types of feature distances in the case of dependency parsing, as suggested by Lin et al., 2019. These features are geographic, genealogical, and syntactic distance.

In order to choose languages for these three features, we query the URIEL database with the lang2vec tool (Littell et al., 2017).

Language branch	Language	Treebank	Train	Dev	Test
Finnic	Estonian (est)	EDT	291k	37.6k	41.4k
	Finnish (fin)	TDT	139.6k	15.7k	18k
	Karelian (krl)	KKPP	-	-	2476
	Livvi (olo)	KKPP	117	-	1075
Mordvin	Erzya (myv)	JR	-	-	12.4k
	Moksha (mdf)	JR	-	-	437
Permic	Komi Permyak (koi)	UH	-	-	320
	Komi Zyrian (kpv)	Lattice	-	-	1804
Sámi	North Sámi (sme)	Giella	14k	-	8700
	Skolt Sámi (sms)	Giellagas	-	-	263
Ugric	Hungarian (hun)	Szeged	17.5k	10k	9097

Table 3.1: Number of tokens in the treebanks used for the Uralic languages.

3.3.1 Geographic Distance

The geographic distance feature is the physical distance between areas where the languages in question are spoken. Of course, languages spoken in close geographic proximity may not necessarily be closely related; however, this feature is useful as such languages will have a history of contact in which they have influenced each other to varying degrees.

In the URIEL database, this feature is derived from data from Glottolog (Hammarström et al., 2020), the World Atlas of Language Structures (WALS) (Dryer and Haspelmath, 2013), and Syntactic Structures of the World's Languages (SSWL) (Collins and Kayne, 2011).

As our geographically proximate transfer languages, we will use Norwegian and Russian. Table 3.2 shows the results of querying the URIEL database for the geographic distance between our eleven Uralic languages and the transfer languages of Norwegian and Russian. As we can see, lower values indicate a higher proximity, and the geographic proximity amongst the Uralic target languages is reflected in their values. Similarly, Norwegian and Russian are shown to be geographically close to the target languages.

Norwegian and Russian provide one other benefit as transfer languages; the former uses the Latin alphabet, while the latter uses the Cyrillic alphabet, and both of these languages are used by our target languages.

Russian may be especially useful for the four of our target languages that use the Cyrillic alphabet (Komi Zyrian, Komi Permyak, Moksha, and Erzya) seeing as the five Uralic languages with training data (Finnish, Estonian, Hungarian, North Sámi, and Livvi) all use the Latin alphabet.

3.3.2 Genealogical Distance

Given that the Uralic languages are part of the same language family, they are the languages with the closest genealogical relatedness to each other. The Uralic languages in UD that have training data, i.e. Estonian, Finnish, Hungarian, Livvi, and North Sámi, will therefore act as transfer languages for each other and the remaining Uralic languages.

	est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun	nor	rus
est	0	0	0	0	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0	0.2
fin		0	0	0	0.1	0.1	0.1	0.1	0	0	0.1	0	0.1
krl			0	0	0.1	0.1	0.1	0.1	0	0	0.1	0.1	0.1
olo				0	0.1	0.1	0.1	0.1	0	0	0.1	0.1	0.1
mdf					0	0	0	0.1	0.1	0.1	0.1	0.1	0.1
myv						0	0	0.1	0.1	0.1	0.1	0.1	0.1
koi							0	0	0.1	0.1	0.1	0.1	0.1
kpv								0	0.1	0.1	0.1	0.1	0.1
sme									0	0	0.1	0.1	0.2
sms										0	0.1	0.1	0.1
hun											0	0.1	0.2

Table 3.2: Geographic distance between the target languages and the first two transfer languages, Norwegian and Russian.

	est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
est	0	0.5	0.33	0.33	0.66	0.66	0.66	0.75	0.66	0.66	0.5
fin		0	0.5	0.5	0.75	0.75	0.75	0.75	0.75	0.75	0.75
krl			0	0.25	0.75	0.66	0.75	0.75	0.75	0.75	0.5
olo				0	0.75	0.66	0.75	0.75	0.75	0.75	0.5
mdf					0	0.33	0.75	0.75	0.66	0.66	0.5
myv						0	0.66	0.75	0.66	0.66	0.5
koi							0	0.25	0.75	0.75	0.5
kpv								0	0.75	0.75	0.5
sme									0	0.6	0.5
sms										0	0.5
hun											0

Table 3.3: Genealogical distance between the Uralic languages.

Table 3.3 shows the genealogical distances between each of the Uralic languages in our study. As we can see, lower values indicate a closer genealogical distance. Several values appear quite high, e.g. 0.75 for Finnish and Moksha. However, this is to be expected, as our eleven Uralic languages are not equally closely related to each other. Referring back to Table 3.1, we can see the language genera, which indicate which of the languages are most closely related. For example, Finnish is far more closely related to the other Finnic languages of Estonian, Karelian, and Livvi, than to the Mordvin language of Moksha. These relationships are reflected in the syntactic distances in Table 3.3.

3.3.3 Syntactic Distance

According to Lin et al., 2019, syntactic distance is significantly less important in a parsing context than geographic and genealogical distance. In order to keep the amount of time spent training models reasonable, we are therefore only considering syntactically similar transfer languages for a few of the target languages in order to see if they have a significant effect on results.

In order to choose which syntactically similar transfer languages to use, we looked at the syntactic distance between the transfer languages and a wide selection of syntactically diverse languages. The distances between the Uralic target languages

	est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
est	0	0.41	0.66	0.66	0.66	0.49	0.61	0.42	0.47	0.66	0.48
fin		0	0.57	0.57	0.57	0.6	0.65	0.52	0.55	0.57	0.41
krl			0	0	0	0.73	0.75	0.72	0.74	0	0.55
olo				0	0	0.73	0.75	0.72	0.74	0	0.55
mdf					0	0.73	0.75	0.72	0.74	0	0.55
myv						0	0.49	0.61	0.35	0.73	0.59
koi							0	0.66	0.47	0.75	0.64
kpv								0	0.55	0.72	0.6
sme									0	0.74	0.58
sms										0	0.55
hun											0

Table 3.4: Syntactic distance between the Uralic languages.

	est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
afr	0.70	0.65	0.71	0.71	0.71	0.78	0.80	0.75	0.78	0.71	0.67
arb	0.62	0.62	0.67	0.67	0.67	0.74	0.75	0.68	0.73	0.67	0.69
dan	0.43	0.52	0.66	0.66	0.66	0.52	0.60	0.58	0.55	0.66	0.50
eng	0.54	0.53	0.57	0.57	0.57	0.72	0.78	0.65	0.71	0.57	0.60
eus	0.64	0.62	0.53	0.53	0.53	0.76	0.76	0.72	0.75	0.53	0.55
deu	0.61	0.54	0.58	0.58	0.58	0.65	0.68	0.72	0.64	0.58	0.54
gle	0.63	0.61	0.63	0.63	0.63	0.80	0.81	0.71	0.79	0.63	0.66
grc	0.72	0.67	0.69	0.69	0.69	0.79	0.85	0.77	0.79	0.69	0.71
heb	0.62	0.49	0.54	0.54	0.54	0.68	0.73	0.72	0.70	0.54	0.52
hin	0.58	0.54	0.58	0.58	0.58	0.68	0.65	0.66	0.64	0.58	0.54
hye	0.59	0.56	0.54	0.54	0.54	0.57	0.61	0.69	0.61	0.54	0.43
jpn	0.72	0.67	0.59	0.59	0.59	0.76	0.73	0.79	0.72	0.59	0.57
nld	0.60	0.61	0.63	0.63	0.63	0.67	0.71	0.74	0.66	0.63	0.59
spa	0.55	0.47	0.56	0.56	0.56	0.70	0.81	0.68	0.74	0.56	0.54
swe	0.54	0.53	0.64	0.64	0.64	0.63	0.70	0.67	0.65	0.64	0.63

Table 3.5: Syntactic distance between the Uralic languages and a selection of other languages with varied syntax: Afrikaans, Arabic, Danish, English, Basque, German, Irish, Ancient Greek, Hebrew, Hindi, Armenian, Japanese, Dutch, Spanish, and Swedish.

	SVO	SOV	VSO	VOS	OVS	OSV	SV	VS	VO	OV
est	1	0	0	0	0	0	1	0	1	0
fin	1	0	0	0	0	0	1	0.33	1	0
hun	1	0.67	0.33	0	0	0	1	0.33	1	0.67
dan	1	0	0	0	0	0	1	0	1	0
spa	1	0	0	0	0	0	1	0.67	1	0
hye	1	0.67	0.33	0.33	0.33	0.33	1	0.33	1	0.67

Table 3.6: Use of basic word order and subject/object placement before or after the verb in Estonian, Finnish, and Hungarian and their chosen transfer languages Danish, Spanish, and Armenian.

themselves can be seen in Table 3.4, while the distances between the Uralic target languages and potential transfer languages can be seen in Table 3.5.

These distances come from the URIEL database (Littell et al., 2017), and are calculated based on information from WALS (Dryer and Haspelmath, 2013), SSWL (Collins and Kayne, 2011), and Ethnologue (Lewis et al., 2015). We can make a few observations from these values, in which lower scores indicate a closer distance between a pair of languages.

Firstly, many of the Uralic languages are less syntactically similar than one might expect. While some are extremely similar in this sense, such as Karelian and Livvi, others, like Moksha and Komi Permyak, are more diverse.

Secondly, very few of the potential transfer languages have distance scores under 0.5 for any of the target languages, with many of them being significantly higher. Three notably low scores are found for the language pairs Estonian-Danish, Finnish-Spanish, and Hungarian-Armenian, however, at 0.43, 0.47, and 0.43, respectively.

It is worth noting that the distance between Russian and Finnish was lower, at 0.38, but since we are already using Russian as one of our geographically similar transfer languages, we chose the next most similar candidate. This way, we can try to observe the different effects of adding a geographically proximate vs syntactically similar transfer language.

We will therefore consider these three language pairs and see whether syntactic similarity can further boost parsing alongside geographic and genealogical similarities in our models.

In order to take a closer look at what these syntactic distance scores mean in the case of these language pairs, we can compare their word order. Table 3.6 contains information about basic word order, i.e. Subject-Object-Verb (SVO), SOV, VSO, VOS, OVS, and/or OSV, as well as whether the subject and object come before or after the verb in a sentence.

We can see that our chosen syntactically similar transfer languages are very similar to the three target languages. Estonian and Danish share the same values here, while Finnish and Spanish differ only in a slightly higher occurrence of the verb coming before the subject in Spanish. Hungarian and Armenian are also very similar, except for the use of object-before-subject syntax found in Armenian, but not in Hungarian.

By considering these syntactically similar transfer languages, we ultimately want to see what sort of effect they have on results as compared to the geographically proximate and genealogically related transfer languages, recalling that Lin et al., 2019, found syntactic similarity to be the least important for boosting parsing performance out of those three features. Therefore, we chose our syntactically similar transfer languages based on one other factor: their high genealogical and geographical distances from

the target languages. In this way, we can get a better idea of the different effects these three features can have on parsing results.

3.4 Experimental Setup

The three types of language distances we are focusing on are based on findings from Lin et al., 2019; in that paper, the researchers limited their experiments to zero-shot transfer models. In order to expand on their research, we will therefore include both zero-shot and other models in this paper.

Thus, we establish four distinct types of parsing models on which to test target language data.

- The first type is the target-trained model; this is a single-source model simply trained on the target data itself. Given that training data is only available for a subset of the languages, this type of model cannot be trained for every single target language.
- The second type is a single-source zero-shot transfer model; in other words, here we train a single-source model on data from one transfer language, and test this model with target data.
- The third type is a multi-source zero-shot transfer model.
- The fourth and final type is a cross-lingual model; this model is also multi-source, but models of this type are trained on various combinations of data from both the target and transfer languages.

3.4.1 Proxy Treebanks

As many of the models we will be training are zero-shot transfer models, many testing cases will require proxy treebanks.

We will select the proxy treebank based on the highest dev results for each model. This proxy will then be used for each language during zero-shot testing of that model.

This way, we can choose the proxy treebank from any of the treebanks used when training the model. The exception is the treebank for Livvi, which does not have dev data and is not big enough for dev data to be created. This treebank will therefore not be used as a proxy for any testing cases.

3.5 Evaluation Metrics

We evaluate the performance of models on each target language using their labeled attachment scores (LAS). Attachment scores reflect the percentage of words that were assigned the correct head during parsing. LAS only counts instances where the accompanying label was assigned correctly as well.

4 Results and Discussion

4.1 Target Training

Table 4.1 contains the test results of the languages that have training data being tested on single-source target-trained models.

The poor results of the Livvi model can likely be explained by the small amount of training data available, as well as the lack of any dev data. The scores for the remaining four languages are far better as a result of significantly larger datasets.

est	fin	olo	sme	hun
79.59	80.25	2.28	61.83	68.05

Table 4.1: LAS scores for single-source target-trained models.

4.2 Zero-shot Transfer

4.2.1 Single-source Models

As can be seen in Table 4.2, zero-shot transfer does not improve upon our target training scores for any language other than Livvi. For Livvi, the highest LAS in a single-source zero-shot setting is achieved using the Finnish-trained model. At 30.44, this is clearly a significant improvement over the model trained only on Livvi.

For the other languages that have training data, we do not see any improvements, or indeed any scores that come anywhere near the target-trained scores. This is unsurprising, as these treebanks are much larger and provide enough training data to achieve decent scores by themselves. Interestingly, however, their scores here are also far lower than the aforementioned 30.44 reached when testing Livvi.

For the remaining six languages that have no training data, scores are generally low; Karelian achieved a slightly higher score than Livvi using the Finnish model, but other scores are far lower.

	est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
est	-	16.89	13.99	15.73	12.83	9.66	9.27	6.62	8.76	12.15	7.90
fin	17.87	-	34.39	30.44	7.49	9.41	6.02	6.62	7.29	13.08	5.33
olo	0.93	0.68	1.16	-	2.14	2.46	3.01	2.30	1.11	1.56	0.76
hun	7.70	6.76	7.18	9.21	11.41	8.91	6.77	8.09	4.84	13.71	-
sme	7.55	7.89	6.11	7.33	11.23	8.94	8.52	9.19	-	18.69	4.61
nor (nynorsk)	7.74	7.85	8.31	7.53	5.53	6.42	4.01	5.24	7.41	13.40	7.43
rus (taiga)	5.34	5.48	4.59	5.24	19.79	12.68	15.79	12.82	3.35	8.10	4.43

Table 4.2: LAS scores for single-source zero-shot transfer models.

		est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
geo + geo	*nor* + rus	6.82	7.01	6.14	10.15	18.18	13.77	15.54	14.43	5.30	7.48	7.39
geo + gen	*est* + nor	-	19.54	14.77	18.01	9.63	9.91	10.28	9.47	9.12	13.71	8.60
	est + rus	-	18.92	14.74	18.21	22.46	16.43	22.06	15.07	9.26	13.08	8.90
	fin + nor	19.67	-	37.46	32.26	9.45	9.54	7.52	6.16	8.24	19.00	5.71
	fin + rus	20.26	-	36.49	31.92	19.96	14.54	15.54	12.09	7.60	13.71	7.11
	hun + *nor*	8.02	8.32	7.85	12.16	14.62	12.78	15.79	9.83	5.84	12.77	-
	hun + rus	9.16	8.84	9.11	12.23	21.57	15.32	21.80	14.43	5.56	13.40	-
	nor + olo	4.13	5.51	4.75	-	5.88	6.87	5.26	5.97	5.56	8.10	6.60
	nor + *sme*	8.20	9.29	7.14	7.06	4.46	5.81	8.02	4.00	-	14.64	5.28
	olo + *rus*	7.74	10.10	10.57	-	18.54	14.22	19.55	13.37	4.79	14.33	5.19
	rus + *sme*	9.96	9.44	9.47	8.74	20.68	14.24	16.29	14.34	-	18.69	7.19
gen + gen	*est* + fin	-	-	32.90	30.31	13.90	11.61	9.02	7.81	10.63	14.33	8.57
	est + olo	-	19.11	14.90	-	7.49	7.07	4.01	9.15	9.16	13.08	7.86
	est + hun	-	17.26	13.74	17.41	11.05	11.12	7.77	7.77	9.33	13.08	-
	est + sme	-	18.48	14.45	16.87	11.59	8.92	8.77	9.01	-	17.13	7.80
	fin + olo	19.01	-	38.24	-	8.38	9.57	6.77	5.24	8.42	15.58	5.41
	fin + hun	19.60	-	38.78	30.91	10.16	9.92	7.02	5.74	7.64	21.18	-
	fin + sme	17.55	-	40.66	28.23	6.77	7.49	5.76	7.12	-	17.45	5.71
	hun + olo	9.46	9.19	11.05	-	10.34	8.94	11.78	8.04	4.70	15.26	-
	hun + sme	10.53	11.22	9.76	12.50	8.91	9.15	10.78	8.13	-	22.74	-
	olo + *sme*	7.51	9.14	9.05	-	7.66	6.54	5.01	4.55	-	21.81	4.74

Table 4.3: LAS scores for multi-source zero-shot transfer models trained on two languages. The language distance each transfer language choice was based on are shown in the first column; these are geographical (geo) or genealogical (gen). Proxy treebanks used for each model are italicized in the second column. Top results for each treebank are highlighted.

4.2.2 Multi-source Models

In Tables 4.3 and 4.4, we can see the results from the multi-source zero-shot transfer models.

The majority of the highest-performing models are trained on a combination of geographically proximate and genealogically related languages. Those trained only on Norwegian and Russian did not perform particularly highly for any languages, while many of those trained only on genealogically related languages performed quite well.

For models trained on three different treebanks, most of the highest results were reached by those trained on one geographically proximate transfer language, and two genealogically related ones, rather than vice versa.

Furthermore, the top results reached by models trained on three treebanks are higher than those trained on two.

As in the single-source models, the languages that have training data available do not see any improvement when parsed with these models as compared to target-trained ones. Again, the exception is Livvi, for which every model performed better.

On the other hand, the remaining languages achieve varying levels of success. Karelian reaches its highest LAS of 40.92 with the model trained on Finnish, Hungarian, and Livvi, and has similar results for several other models.

Meanwhile, Hungarian does not achieve a single LAS over 10 in this set of models. Referring back to 3.2, and recalling that geographic distance was found by Lin et al., 2019, to be the single most important distance feature when selecting transfer languages for dependency parsing, this could be partially explained by the geographic distance of these languages. Karelian is geographically very close to many of the training languages, while Hungarian is more geographically distant from all of them.

Moksha, Erzya, and Komi Zyrian achieve their highest LAS score on the same model trained on Estonian, Livvi, and Russian. Here we can see how Russian is especially useful as a transfer language for these LR languages; as relatively small languages

		est	fin	krl	olo	mdf	myv	koi	kpv	sme	sms	hun
2 geo	*est* + nor + rus	-	18.90	14.48	17.34	21.03	16.96	20.55	14.66	9.64	12.15	9.49
+ 1 gen	*fin* + nor + rus	18.91	-	34.13	28.63	20.68	15.68	17.29	12.68	8.34	14.95	6.79
	hun + *nor* + rus	8.56	10.23	7.79	12.10	21.21	15.06	15.79	16.13	5.79	14.95	-
	nor + olo + rus	7.23	8.98	6.88	-	21.93	15.05	18.55	13.79	6.37	9.03	6.52
	nor + rus + *sme*	9.61	10.74	8.14	8.00	17.47	12.62	16.29	12.45	-	19.00	6.24
1 geo	*est* + fin + nor	-	-	31.48	28.36	8.73	9.24	10.28	7.63	9.01	15.89	7.47
+ 2 gen	*est* + fin + rus	-	-	32.58	29.70	19.96	14.14	17.54	14.15	10.10	14.02	8.41
	est + hun + nor	-	18.32	15.35	19.15	11.05	10.65	8.27	7.90	10.21	13.08	-
	est + hun + rus	-	18.96	14.61	17.47	20.86	14.99	15.29	14.75	9.90	16.51	-
	est + nor + olo	-	19.02	15.19	-	13.90	11.29	9.27	8.82	10.78	12.77	8.83
	est + nor + sme	-	18.28	14.32	17.20	10.70	10.09	8.27	9.60	-	19.00	8.82
	est + olo + rus	-	19.70	14.77	-	23.53	17.37	21.55	15.90	9.10	16.20	9.17
	est + rus + sme	-	18.64	14.67	17.81	22.82	15.13	19.55	14.89	-	17.76	8.57
	fin + hun + nor	19.24	-	34.58	32.06	11.76	10.15	10.53	7.49	6.67	17.13	-
	fin + hun + rus	18.95	-	37.36	30.31	21.75	13.98	13.53	11.90	7.49	16.20	-
	fin + nor + olo	20.32	-	37.88	-	10.70	9.53	6.77	7.17	8.31	11.21	6.08
	fin + nor + sme	18.16	-	37.52	28.16	10.34	11.77	5.76	6.66	-	16.51	4.82
	fin + olo + rus	18.75	-	38.24	-	20.14	13.75	15.29	13.88	7.94	15.58	5.02
	fin + rus + sme	18.07	-	35.97	27.96	21.39	15.21	13.28	13.88	-	16.82	6.37
	hun + nor + olo	9.56	10.54	8.69	-	11.05	10.20	15.29	9.93	5.00	15.26	-
	hun + nor + *sme*	8.56	9.47	5.79	7.19	6.95	9.20	6.02	5.93	-	19.63	-
	hun + olo + rus	9.59	9.03	9.41	-	21.57	14.00	17.54	13.47	4.64	20.56	-
	hun + rus + *sme*	9.86	10.33	7.72	9.54	20.50	14.87	22.31	14.61	-	20.25	-
	nor + olo + *sme*	9.31	10.11	8.18	-	6.24	5.97	6.77	4.78	-	18.69	4.64
	olo + rus + *sme*	9.49	10.59	9.99	-	19.61	14.73	20.30	14.06	-	20.25	6.26
gen	*est* + fin + hun	-	-	34.13	31.72	13.73	11.01	11.53	9.56	10.29	12.77	-
	est + fin + olo	-	-	32.90	-	8.20	9.32	6.02	7.03	10.19	16.51	9.64
	est + fin + sme	-	-	32.68	30.85	11.41	11.08	10.28	7.40	-	18.69	9.14
	est + hun + sme	-	18.24	14.32	16.13	10.34	10.26	9.27	7.40	-	14.95	-
	est + hun + olo	-	17.60	12.86	-	9.63	9.51	9.27	8.13	9.26	20.87	-
	est + olo + sme	-	19.58	15.32	-	11.05	10.11	10.78	9.83	-	18.38	8.35
	fin + hun + olo	20.13	-	40.92	-	7.49	9.27	9.02	7.12	8.50	19.00	-
	fin + hun + sme	20.34	-	37.88	30.58	6.24	7.92	5.01	6.66	-	23.99	-
	fin + olo + sme	17.51	-	37.43	-	10.34	9.72	6.52	6.20	-	19.00	6.18
	hun + olo + *sme*	10.29	10.54	9.02	-	7.31	7.48	7.52	5.51	-	24.61	-

Table 4.4: LAS scores for multi-source zero-shot transfer models trained on three languages. The language distance each transfer language choice was based on are shown in the first column; these are geographical (geo) or genealogical (gen). Proxy treebanks used for each model are in italicized in the second column. Top results for each treebank are highlighted.

that have historically had a lot of close contact with Russian, they demonstrate why geographic distance is such an important feature to consider.

The same model was close to providing the highest score for Komi Permyak as well. The actual top model for this language was trained on Hungarian, Russian, and North Sámi.

Looking at the results for Moksha, Erzya, Komi Permyak, and Komi Zyrian, there is a clear pattern of Russian being especially useful as a transfer language for these four target languages. Like Russian, these languages are written in Cyrillic, suggesting a shared script in target and transfer languages improve results as well. This can be explained by the use of a character model by UUParser, which learns character-level encoding from the training data. At this level, a character from a transfer language in a different script than that of the target language is not as useful as one using the same script.

For Skolt Sámi, the best-performing model was trained on Hungarian, Livvi, and North Sámi. Interestingly, neither Russian or Norwegian are part of this best-performing model. However, Hungarian, Livvi, and North Sámi are all have low distances from Skolt Sámi in all three ways we have discussed.

		est	fin	hun	olo	sme
	target	79.59	80.25	68.05	2.28	61.83
geo	target + nor	**79.83**	80.23	**69.03**	**12.10**	**62.21**
	target + rus	**79.82**	**80.30**	**69.34**	**20.56**	**63.65**
geo + gen	target + est + nor	-	**80.27**	**68.69**	**17.74**	60.85
	target + est + rus	-	**80.36**	66.96	**21.03**	59.70
	target + fin + nor	**79.65**	-	**68.47**	**33.00**	**61.85**
	target + fin + rus	79.36	-	**69.06**	**32.73**	**62.09**
	target + hun + nor	**80.03**	79.85	-	**15.59**	**62.37**
	target + hun + rus	**79.90**	80.09	-	**17.81**	61.18
	target + nor + olo	**80.05**	**80.36**	**69.08**	-	61.45
	target + nor + rus	**79.76**	79.23	**68.62**	**13.58**	60.92
	target + nor + sme	**79.63**	79.86	67.55	**11.63**	-
	target + olo + rus	**79.99**	80.25	**69.73**	-	**62.22**
	target + rus + sme	**79.74**	79.92	**69.53**	**15.39**	-
gen	target + est	-	79.97	68.61	**20.90**	45.15
	target + fin	**80.29**	-	**69.93**	**33.94**	**62.81**
	target + hun	**79.65**	**80.35**	-	**14.99**	**62.28**
	target + olo	**80.10**	**80.45**	**69.13**	-	**62.67**
	target + sme	**79.79**	79.86	**69.07**	**12.57**	-
	target + est + fin	-	-	64.02	**38.44**	60.49
	target + est + hun	-	**80.30**	-	**22.65**	60.46
	target + est + olo	-	38.44	**69.11**	-	60.89
	target + est + sme	-	79.81	**68.86**	**20.16**	-
	target + fin + hun	79.50	-	-	**35.95**	61.67
	target + fin + olo	33.33	-	**68.82**	-	**68.66**
	target + fin + sme	**79.94**	-	**68.20**	**35.82**	-
	target + hun + olo	**79.80**	**80.36**	-	-	60.47
	target + hun + sme	79.42	79.86	-	**17.81**	-
	target + olo + sme	**80.24**	80.12	**68.48**	-	-

Table 4.5: LAS scores for cross-lingual models. The language distance each transfer language choice was based on are shown in the first column; these are geographical (geo) or genealogical (gen). Scores that are higher than the target-trained baselines are in bold. Top results for each treebank are highlighted.

4.3 Cross-lingual Models

As we can see in Table 4.5, cross-lingual models tend to produce significantly higher scores than zero-shot transfer models do. Here, we see that many scores for each of the five languages that have training data are higher than the initial single-source target-trained models.

For Estonian, the best model was trained on a cross-lingual model with Finnish, while Finnish achieved the highest results on a cross-lingual model with Livvi. As we have seen in Chapter 3, the languages in both of these pairs have low distances geographically, genealogically, and syntactically, and these low distances are reflected in their high LAS scores in these models.

Hungarian performed best on a cross-lingual model with Finnish. Here too, we can see the low distance between the two languages; interestingly, Finnish is the least genealogically related language to Hungarian (see Table 3.3). However, they are still genealogically related, and very geographically proximate and syntactically similar.

		est	fin	hun
Top x-ling model	est + fin	80.29		
	fin + olo		80.45	
	fin + hun			69.93
Target + synt.	est + dan	79.70		
	fin + spa		78.54	
	hun + hye			69.98
Top x-ling model + synt.	dan + est + fin	79.27		
	fin + olo + spa		78.56	
	fin + hun + hye			69.10

Table 4.6: LAS results for models including the selected syntactically similar transfer languages (abbreviated as *synt.*) for Estonian, Finnish, and Hungarian. The treebanks used are Danish DDT, Spanish Ancora, and Armenian ArmTDP.

While Estonian, Hungarian, and Finnish did achieve good results with many models, the improvement from the baseline single-source target-trained models is less than 1 LAS point for Estonian and Finnish and just under 2 for Hungarian. The difference is much more significant for North Sámi and Livvi.

North Sámi improved on its baseline by 6.83 LAS points using the model trained on Finnish, Livvi, and North Sámi.

Livvi only achieved a high LAS of 38.44 with the model trained on Estonian, Finnish, and Livvi; however, this is a significant improvement over its baseline of 2.28.

As none of the highest-scoring cross-lingual models were trained with Norwegian or Russian, it is difficult to say how much geographic distance in transfer languages improved results versus genealogical relatedness. Instead, the best-performing transfer languages are similar in both regards. There is no clear pattern suggesting one is better than the other, as only a few cross-lingual models performed significantly lower than the baselines.

These results do suggest, however, that in cases where target language data is available, the need for geographically proximate transfer languages decreases. While many of the models including data from geographically proximate languages outperform baselines, they do not outperform the highest-scoring models trained only on target data and genealogically related transfer languages. This is in contrast to the results for the target languages without training data in Tables 4.3 and 4.4, where geographically proximate transfer languages seem to boost results significantly.

It is also worth noting that the Finnish and Estonian treebanks, which were used to train many high-performing models, have significantly more data than the rest. This may have increased their performance further.

4.4 Adding Syntactically Similar Transfer Languages

In Table 4.6, we can see the results of adding our chosen syntactically similar languages to Estonian, Finnish, and Hungarian.

As single-source zero-shot transfer models, these transfer languages were not useful; LAS scores for all three were below 7.

In the case of Estonian and Finnish, the best performing cross-lingual model from 4.5 outperformed all models using their corresponding syntactically similar transfer languages.

For Estonian, the best performing cross-lingual model configuration plus Danish did slightly worse than the model trained on Estonian and Danish alone; for Finnish,

the reverse is true, i.e. the model trained on three languages did a little better than the one trained on two. However, in each of these cases, the difference is not significant.

In the case of Hungarian, the model trained on Hungarian and Armenian did slightly outperform the best performing cross-lingual model by 0.05 LAS points. This could be because Hungarian is the most geographically remote of the Uralic languages, and therefore might not have benefited as much as the others from our chosen geographically proximate transfer languages.

5 Conclusion and Future Work

We explored different options for the multilingual parsing of Uralic languages, considering different types of language distances and parser model configurations.

We found that geographic proximity and genealogical relatedness in transfer languages appear to boost scores in most cases, though note that it is difficult to tell how much can be attributed to either individual feature as the chosen genealogically related languages also have high geographic proximity.

However, we found that most highest-scoring multi-source zero-shot models were those trained on both genealogically related and geographically proximate transfer languages, as most included Norwegian or Russian as well as Uralic languages. For the highest-scoring models trained on three languages, the majority were trained on either Norwegian or Russian and two Uralic languages, rather than just one Uralic language and both geographically proximate languages. For the highest-scoring cross-lingual models, on the other hand, all highest-scoring ones were trained only on the target language plus one or two other Uralic languages.

Syntactic similarity was suggested to be significantly less important than the other two features, according to Lin et al., 2019, and so we only considered syntactically similar transfer languages for three target languages and only trained a few models with these. Here, we found that these models generally performed slightly worse than others.

The notable exception was Hungarian, which performed better when a syntactically similar language was added to a cross-lingual model.

For the remaining languages with training data, we found cross-lingual models performed the best by far, and specifically ones trained on Uralic languages only.

For the languages without training data, we had to rely on zero-shot transfer, which did not perform as well. The highest scores for these were all in the 20's, with the exception of Karelian which reached an LAS of 40.92 with a model trained on Finnish, Hungarian, and Livvi.

In future work, it would be interesting to further explore the effect of geographical distance vs genealogical relatedness by using transfer languages that have just one of those features; however, this could be difficult to manage, as most genealogically related languages tend to be geographically proximate too.

As Hungarian is the most geographically remote language out of the group of Uralic languages we included, future work could also consider languages which are more geographically proximate to Hungarian specifically. Looking at more syntactically similar languages, perhaps with varying degrees of distance from the transfer languages, could shed further light on how this type of distance affects parsing.

Since the Livvi treebank was so small, we could not create a dev set for it, which affected the optimization process. This also meant we decided not to choose it as a proxy treebank for any model because we did not have any dev results. Furthermore, the split of the treebank into training and test sets is very uneven, with a small training set and a much larger test set. This could be better balanced in future work, preferably with a dev set included as well.

As the North Sámi treebank does not include a dev set, we used the UUParser flag to generate one. As this flag picks a random set of sentences each time it is used, each model trained on North Sámi has a slightly different (though equally sized) training

set. It is therefore difficult to say how much this method affected the results, and in future work it would be useful to have a dev set rather than using the flag.

Previous work has shown that the increase in performance of multi-source models, as compared to single-source ones, is the most dramatically obvious for LR languages. This could be seen in our results for Karelian, for example. In future work, it would be interesting to try to limit the sizes of the larger Uralic treebanks in order to simulate an LR scenario for them, making them as small as the genuinely LR ones.

This would also be useful as varying sizes of the treebanks used played a role in the performance of the models, and it is difficult to say to what extent. Additionally, it would be interesting to investigate what role the genre of the treebanks may have played.

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